

Rocksdb_dataset-8

Commit 141:

8e16f8fecf469f2068f1020008993c9a4b6fceb6

The 1st symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2024947.818645833
4  The new version is executed by default
5  2039774.280061225
6  The new version
7  2036898.633666667
8  Comparison of New and old versions (along symbolic execution path)
9  -0.005867162372886928
10 Comparison of New and old versions (along the default execution path)
11 0.0014117768783523128
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2061873.7268421054
```

```
4 The new version is executed by default
5 2050390.6473888888
6 The new version
7 2059583.3791578945
8 Comparison of New and old versions (along symbolic execution path)
9 0.0011120441674700849
10 Comparison of New and old versions (along the default execution path)
11 -0.004463393840731234
```

No performance regression

The 3rd symbolic execution

Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2037109.0387916667
4 The new version is executed by default
5 2036835.096083333
6 The new version
7 2030813.38
8 Comparison of New and old versions (along symbolic execution path)
9 0.003100067615108411
10 Comparison of New and old versions (along the default execution path)
11 0.0029651745170859224
```

No performance regression

Commit 142:

c5921df3d71c723ae4ca5650ff1f65305a53e5e0

The 1st symbolic execution

Testcase:for(int i =0; i < 10; i++){rocksdb::Slice start_key("key_exit:300"), end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db->DefaultColumnFamily(), start_key, end_key);rocksdb::Slice

```
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1676816.7943404254
4  The new version is executed by default
5  1662675.3088936172
6  The new version
7  1682693.0460833337
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0034921709319390925
10 Comparison of New and old versions (along the default execution path)
11 -0.011896250023918545
```

No performance regression

The 2nd symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2028786.7272499998
4  The new version is executed by default
5  2047838.3960416664
6  The new version
7  2033974.0331666667
8  Comparison of New and old versions (along symbolic execution path)
9  -0.002550330452641491
10 Comparison of New and old versions (along the default execution path)
11 0.006816391285691294
```

No performance regression

Commit 143:

ca7367a00319b333e783cab60cb600dd54f18b8f

The 1st symbolic execution

```
Testcase:for(int i=0;i < 10;i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1669880.3705102045
4  The new version is executed by default
5  1693581.9029375
6  The new version
7  1678878.5215833334
8  Comparison of New and old versions (along symbolic execution path)
9  -0.005359620102020753
10 Comparison of New and old versions (along the default execution path)
11 0.0087578589904766
```

No performance regression

The 2nd symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
```

```
2 The new version executes along symbols
3 2039917.0731489358
4 The new version is executed by default
5 2038889.5685
6 The new version
7 2038093.3580000005
8 Comparison of New and old versions (along symbolic execution path)
9 0.0008948143331004992
10 Comparison of New and old versions (along the default execution path)
11 0.0003906643907523979
```

No performance regression

The 3rd symbolic execution

Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2046245.0366041667
4 The new version is executed by default
5 2042766.2005833331
6 The new version
7 2032719.7060625001
8 Comparison of New and old versions (along symbolic execution path)
9 0.006653809918469268
10 Comparison of New and old versions (along the default execution path)
11 0.00494239047856415
```

No performance regression

The 4th symbolic execution

Testcase:for(int i=0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db->DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice

```
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db->CompactRange(comp_options, &start_key1, &end_key2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1700765.3665208332
4  The new version is executed by default
5  1694720.8502083335
6  The new version
7  1671008.1308125
8  Comparison of New and old versions (along symbolic execution path)
9  0.017807953869060086
10 Comparison of New and old versions (along the default execution path)
11 0.01419066667515474
```

No performance regression

The 5th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1676207.216125
4  The new version is executed by default
5  1695424.8967659576
6  The new version
7  1668673.7910625
8  Comparison of New and old versions (along symbolic execution path)
9  0.004514618197306951
```

```
10 Comparison of New and old versions (along the default execution path)
11 0.016031357265115415
```

No performance regression

The 6th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Compilation error occurred, terminated.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2037115.6097291664
4 The new version is executed by default
5 2037578.5965510206
6 The new version
7 2048254.9309166668
8 Comparison of New and old versions (along symbolic execution path)
9 -0.005438444706936521
10 Comparison of New and old versions (along the default execution path)
11 -0.005212405059788179
```

No performance regression

The 7th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2035078.1428958334
4 The new version is executed by default
5 2040046.2856458332
6 The new version
```

```
7 2043778.4631249998
8 Comparison of New and old versions (along symbolic execution path)
9 -0.004256978134441912
10 Comparison of New and old versions (along the default execution path)
11 -0.0018261164536688208
```

No performance regression

The 8th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 1676043.2297708334
4 The new version is executed by default
5 1836323.2583200003
6 The new version
7 1663749.2356666669
8 Comparison of New and old versions (along symbolic execution path)
9 0.007389331180813611
10 Comparison of New and old versions (along the default execution path)
11 0.1037259816285852
```

There may be performance regression

The 9th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```


Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1679136.1340416663
4  The new version is executed by default
5  1669198.5630638301
6  The new version
7  1688236.1024782606
8  Comparison of New and old versions (along symbolic execution path)
9  -0.005390222625399361
10 Comparison of New and old versions (along the default execution path)
11 -0.011276585891324195
```

No performance regression

The 10th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1689599.5515918366
4  The new version is executed by default
5  1688560.2762708336
6  The new version
7  1660363.0214375
8  Comparison of New and old versions (along symbolic execution path)
9  0.017608516798347188
10 Comparison of New and old versions (along the default execution path)
11 0.016982584211566697
```

No performance regression

The 11th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1658751.3574255318
4  The new version is executed by default
5  1898289.0942244898
6  The new version
7  1663060.7553958334
8  Comparison of New and old versions (along symbolic execution path)
9  -0.002591245062045772
10 Comparison of New and old versions (along the default execution path)
11 0.14144302188928057
```

There may be performance regression

The 12th symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Compilation error occurred, terminated.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2035490.1844680852
4  The new version is executed by default
5  2038043.8880208333
6  The new version
```

```
7 2037409.8128775507
8 Comparison of New and old versions (along symbolic execution path)
9 -0.0009421906173870542
10 Comparison of New and old versions (along the default execution path)
11 0.0003112162998699775
```

No performance regression

The 13th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2039573.2958085109
4 The new version is executed by default
5 2041271.93700000002
6 The new version
7 2040168.2218163265
8 Comparison of New and old versions (along symbolic execution path)
9 -0.0002916063496401215
10 Comparison of New and old versions (along the default execution path)
11 0.000540992243615598
```

No performance regression

The 14th symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2031066.9901224487
```

```
4 The new version is executed by default
5 2042056.5525306123
6 The new version
7 2025885.8559999994
8 Comparison of New and old versions (along symbolic execution path)
9 0.002557465963397894
10 Comparison of New and old versions (along the default execution path)
11 0.00798203733084007
```

No performance regression

The 15th symbolic execution

Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Compilation error occurred, terminated.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2038205.964104167
4 The new version is executed by default
5 2049604.7221666668
6 The new version
7 2047876.0433125
8 Comparison of New and old versions (along symbolic execution path)
9 -0.004722004166175706
10 Comparison of New and old versions (along the default execution path)
11 0.0008441325634976294
```

No performance regression

The 16th symbolic execution

Testcase:for(int i = 0; i < 10; i++){rocksdb::Slice start_key("key_exit:300"), end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db->DefaultColumnFamily(), start_key, end_key);rocksdb::Slice start_key1("key_exit:200");rocksdb::Slice end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db->CompactRange(comp_options, &start_key1, &end_key2);}

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1675969.1169791666
4  The new version is executed by default
5  1674451.648723404
6  The new version
7  1669697.7069791667
8  Comparison of New and old versions (along symbolic execution path)
9  0.0037560152198724715
10 Comparison of New and old versions (along the default execution path)
11 0.0028471870832463947
```

No performance regression

The 17th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1678912.5664166666
4  The new version is executed by default
5  1687419.1155319146
6  The new version
7  1809034.2564599996
8  Comparison of New and old versions (along symbolic execution path)
9  -0.07192881482408245
10 Comparison of New and old versions (along the default execution path)
11 -0.0672265549940812
```

No performance regression

The 18th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1665155.1988124999
4  The new version is executed by default
5  1679039.1927021272
6  The new version
7  1681177.0764468082
8  Comparison of New and old versions (along symbolic execution path)
9  -0.009530154710514355
10 Comparison of New and old versions (along the default execution path)
11 -0.0012716588720085745
```

No performance regression

The 19th symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1666530.0893125
```

```
4 The new version is executed by default
5 1678808.8496170205
6 The new version
7 1677363.2582916666
8 Comparison of New and old versions (along symbolic execution path)
9 -0.006458451337606935
10 Comparison of New and old versions (along the default execution path)
11 0.0008618236498312913
```

No performance regression

Commit 144:

0cc943c06714e7d1fff1df94c5c47f60ace6410e

The 1st symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({"write_buffer_size", "131072"});}

Symbolic execution completed successfully.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 666881.1749999999
4 The new version is executed by default
5 676249.9718510638
6 The new version
7 707859.2880816326
8 Comparison of New and old versions (along symbolic execution path)
9 -0.05789019621779254
10 Comparison of New and old versions (along the default execution path)
11 -0.044654801826833575
```

No performance regression

Commit 145:

6ac13a5f0aecdbd7fd09f01b014f6564b87e0436

The 1st symbolic execution

```
Testcase:for(int i =0;i < 10; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1668768.1345744678
4  The new version is executed by default
5  1719652.5444444444
6  The new version
7  1718046.1353265306
8  Comparison of New and old versions (along symbolic execution path)
9  -0.028682582928832134
10 Comparison of New and old versions (along the default execution path)
11 0.0009350209431998123
```

No performance regression

The 2nd symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2047632.0237142858
4  The new version is executed by default
5  2037540.3691632657
6  The new version
7  2033118.0106666666
8  Comparison of New and old versions (along symbolic execution path)
9  0.007138795176409883
```



```
10 Comparison of New and old versions (along the default execution path)
11 0.002175160749842074
```

No performance regression

Commit 146:

24952ff0883dad4826093411d4573aa00e479772

The 1st symbolic execution

Testcase:for (int i = 0; i < 5000; i++){key = "key_exit" + std::to_string(i + 30001);value = "new_value" + key;db->Put(write_options_sync, key, value);}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 343554.24429166666
4 The new version is executed by default
5 344423.5855208333
6 The new version
7 355575.65069387754
8 Comparison of New and old versions (along symbolic execution path)
9 -0.03380829474333257
10 Comparison of New and old versions (along the default execution path)
11 -0.03136341071522151
```

No performance regression

Commit 147:

17ac19f2c448856f2cfca2d85500f8b0b4af0bdc

Unable to provide test cases

Commit 148:

cc487ba3678b5514994c6357f7a53115f998d909

The 1st symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2033846.3928333335
4  The new version is executed by default
5  2028966.5381041665
6  The new version
7  2036567.7984081635
8  Comparison of New and old versions (along symbolic execution path)
9  -0.0013362705513448364
10 Comparison of New and old versions (along the default execution path)
11 -0.0037323875541675163
```

No performance regression

The 2nd symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2043637.5865918368
4  The new version is executed by default
5  2037886.6740212766
6  The new version
7  2040647.7236874998
8  Comparison of New and old versions (along symbolic execution path)
9  0.001465153867387874
10 Comparison of New and old versions (along the default execution path)
11 -0.0013530261172339107
```

No performance regression

The 3rd symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1705630.1187659574
4  The new version is executed by default
5  1683550.621531915
6  The new version
7  1778751.3255
8  Comparison of New and old versions (along symbolic execution path)
9  -0.04110816710901875
10 Comparison of New and old versions (along the default execution path)
11 -0.05352108673275314
```

No performance regression

The 4th symbolic execution

```
Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db-
>Put(rocksdb::WriteOptions(), key, value2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2030560.5073061225
4  The new version is executed by default
5  2030712.6464791663
6  The new version
7  2027122.5965625
8  Comparison of New and old versions (along symbolic execution path)
```

```
9 0.0016959560065346766
10 Comparison of New and old versions (along the default execution path)
11 0.0017710077933885562
```

No performance regression

The 5th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 1679260.9528723403
4 The new version is executed by default
5 1692714.6314791667
6 The new version
7 1668878.8323958332
8 Comparison of New and old versions (along symbolic execution path)
9 0.0062210151360135605
10 Comparison of New and old versions (along the default execution path)
11 0.01428252226623003
```

No performance regression

The 6th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1696743.7800612245
4  The new version is executed by default
5  1675338.807106383
6  The new version
7  1680555.9100416668
8  Comparison of New and old versions (along symbolic execution path)
9  0.009632449550075545
10 Comparison of New and old versions (along the default execution path)
11 -0.0031043911744385584
```

No performance regression

The 7th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1676355.6360416666
4  The new version is executed by default
5  1677899.8446458336
6  The new version
7  1675514.2779374998
8  Comparison of New and old versions (along symbolic execution path)
9  0.0005021491701058438
10 Comparison of New and old versions (along the default execution path)
11 0.001423781784342859
```

No performance regression

The 8th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1686383.28700000002
4  The new version is executed by default
5  1677271.805255319
6  The new version
7  1662498.045319149
8  Comparison of New and old versions (along symbolic execution path)
9  0.014367079557236992
10 Comparison of New and old versions (along the default execution path)
11 0.008886482590320184
```

No performance regression

The 9th symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice
start_key1("key_exit:200");rocksdb::Slice
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  17034407.40326316
```

```
4 The new version is executed by default
5 16966804.546526317
6 The new version
7 16905293.732333332
8 Comparison of New and old versions (along symbolic execution path)
9 0.007637469834841257
10 Comparison of New and old versions (along the default execution path)
11 0.0036385534121384532
12
```

No performance regression

Commit 149:

0a43d8a261b9c633c0a4e369b1ef33aa5ee32810

The 1st symbolic execution

Testcase:for(int i = 0; i < 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 2035272.4245208334
4 The new version is executed by default
5 2034234.568632653
6 The new version
7 2041408.5606808506
8 Comparison of New and old versions (along symbolic execution path)
9 -0.0030058344410835132
10 Comparison of New and old versions (along the default execution path)
11 -0.0035142362907525913
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i < 10; i++){rocksdb::Slice start_key("key_exit:300"), end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-

```
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice  
start_key1("key_exit:200");rocksdb::Slice  
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-  
>CompactRange(comp_options, &start_key1, &end_key2);}
```

Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:  
2  The new version executes along symbols  
3  1672749.0957083337  
4  The new version is executed by default  
5  1696522.0966734695  
6  The new version  
7  1791428.890122449  
8  Comparison of New and old versions (along symbolic execution path)  
9  -0.06624867728129766  
10 Comparison of New and old versions (along the default execution path)  
11 -0.0529782644302964
```

No performance regression

Commit 150:
2e175124d8851225160a7beb506b6e6a767554e0

Unable to provide test cases

Commit 151:
12829883d7a9b9412217146e20af1e9cb79fc861

The 1st symbolic execution

```
Testcase:for(int i =0;i < 100; i++){rocksdb::Slice start_key("key_exit:300"),  
end_key("key_exit:500");db->DeleteRange(rocksdb::WriteOptions(), db-  
>DefaultColumnFamily(), start_key, end_key);rocksdb::Slice  
start_key1("key_exit:200");rocksdb::Slice  
end_key2("key_exit:700");rocksdb::CompactRangeOptions comp_options;db-  
>CompactRange(comp_options, &start_key1, &end_key2);}
```


Symbolic execution exception occurred, resulting in a timeout.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  1915286.6293469388
4  The new version is executed by default
5  1718586.7758510637
6  The new version
7  1698167.4512244898
8  Comparison of New and old versions (along symbolic execution path)
9  0.12785498742535187
10 Comparison of New and old versions (along the default execution path)
11 0.012024329292055479
```

There may be performance regression

Commit 152: d0374a0a72908fb54bd60e9c32f78e848fd01d69

The 1st symbolic execution

Testcase:for(int i = 0; i< 1000; i++){db->SetOptions({{"write_buffer_size", "131072"}});}

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  712243.1940208334
4  The new version is executed by default
5  688345.5774583333
6  The new version
7  677521.259755102
8  Comparison of New and old versions (along symbolic execution path)
9  0.05124847931455615
10 Comparison of New and old versions (along the default execution path)
11 0.01597635136518658
```

No performance regression

The 2nd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2043376.960959184
4  The new version is executed by default
5  2035753.7012708336
6  The new version
7  2038401.995367347
8  Comparison of New and old versions (along symbolic execution path)
9  0.0024406204483431145
10 Comparison of New and old versions (along the default execution path)
11 -0.001299201091115573
```

No performance regression

The 3rd symbolic execution

Testcase:for(int i = 0; i< 30000; i++){key = "key_missing" + std::to_string(distrib(gen));db->Put(rocksdb::WriteOptions(), key, value2);}

Symbolic execution completed successfully.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  2038152.4692765959
4  The new version is executed by default
5  2035152.6593958333
6  The new version
7  2035438.9093541668
8  Comparison of New and old versions (along symbolic execution path)
9  0.001333157143630526
10 Comparison of New and old versions (along the default execution path)
11 -0.0001406330384164587
```

No performance regression

Commit 153:

7f3ee34cdf705415a9207cd102eb6907ce4f91e5

Unable to provide test cases

Commit 154:

0b815cf3b35182d91f378f6a647f61289c60942f

Unable to provide test cases

Commit 155:

934cf2d40dc77905ec565ffec92bb54689c3199c

Unable to provide test cases

Commit 156:

82794e0a4f3116878db210bce7e3768a68c47173

Unable to provide test cases

Commit 157:

49b0cb64df082978962b1d7ae24693dcb7f9dbf2

Unable to provide test cases

Commit 158:

9072f5db09372f3662a86c884183625a64c4cddd

Non-C file modifications, therefore no performance regression testing is required.

Commit 159:

743a02d6f68811bed8c5f3ce5d7e6d9b6bc18783

Unable to provide test cases

Commit 160:

325dcdf2e54aaaad9ca44938deb2729142e670dc

Non-C file modifications, therefore no performance regression testing is required.